BIRZEIT UNIVERSITY
Department of Electrical and Computer Engineering
ENCS3340 Artificial Intelligence, First Semester, 2025-2026
Machine Learning Project, Due: January 15, 2026
Arabic Sentiment Analysis

1. Project Description.

This project aims to develop and systematically evaluate multiple machine learning models for sentiment classification in Arabic social media data. By emphasizing thorough data analysis, feature engineering, and rigorous comparative evaluation of models, the project seeks to provide a comprehensive understanding of the challenges inherent in Arabic sentiment analysis, as well as the most effective techniques for addressing them.

2. Project Objectives.

The primary objectives of this project are to conduct an initial exploratory analysis of the selected Arabic sentiment dataset to examine its structure, content, and linguistic characteristics, including class distribution. In addition, the project aims to design and evaluate different feature extraction and representation methods. Furthermore, the project involves the implementation and optimization of a diverse set of classification algorithms. Finally, all developed models will be benchmarked using widely accepted evaluation metrics in order to identify the most effective approach for Arabic sentiment classification.

3. Dataset Overview.

The project will utilize the attached Arabic sentiment dataset. The data are annotated into three sentiment categories, namely *Positive*, *Negative*, and *Objective/Neutral*. Notably, the dataset presents several challenges, including class imbalance and the presence of linguistic noise such as spelling variations, informal expressions, and dialect-specific vocabulary, all of which must be carefully addressed during the model development and evaluation stages.

4. Project Tasks and Methodology

The machine learning lifecycle outlines the steps required to develop a predictive model. For this project, students are expected to work on all stages **except data gathering**. Specifically, the tasks and methodology include the following:

1. Perform an initial analysis of the dataset to understand its structure and characteristics, including determining the distribution of sentiment classes.
2. **Data Preprocessing:** Data preprocessing is performed to clean and standardize the textual data prior to feature extraction. This process includes text cleaning operations such as converting all characters to lowercase, removing HTML tags and URLs, and eliminating special characters and numerical tokens. Noise is reduced by correcting common misspellings and expanding contractions to their full

forms. Text normalization is then applied through stemming or lemmatization to reduce words to their canonical forms, along with the removal of stop words that do not contribute meaningful semantic information. In addition, variations in textual representation, such as dates and emojis, are standardized to ensure consistency across the corpus. The selection of specific preprocessing techniques is guided by an analysis of the dataset characteristics and the overall project objectives.

3. **Feature Extraction:** Feature extraction is performed by developing content-based representations that encode the semantic and syntactic properties of tweets, using suitable text representation techniques such as term frequency–inverse document frequency (TF–IDF), word embeddings (e.g., Word2Vec or FastText), or contextual embeddings derived from transformer-based language models. In addition to these automated representations, hand-engineered features are designed to capture linguistic characteristics specific to Arabic text. These include dialectal indicators, negation handling, and character normalization features that unify variant letter forms. Additional handcrafted features may include tweet length, word repetition patterns, punctuation usage, and the presence of emoticons or hashtags. The combination of learned representations and linguistically motivated features enables a richer and more discriminative feature space for downstream classification tasks.
4. **Model Training:** Train and evaluate three classifiers: Decision Trees (including Random Forests), Naïve Bayes, and Neural Networks.
5. **Dataset Splitting and Parameter Optimization:** Split the dataset into 60% training, 20% validation, and 20% testing. Perform parameter optimization for each classifier to enhance performance. Discuss challenges such as class imbalance and dataset sparsity, and propose methods to mitigate these issues, such as resampling, weighting, or other suitable techniques.
6. **Model Evaluation:** Evaluate all trained models on the held-out test set using the following performance metrics: Accuracy, Precision, Recall, F1-Score.

5. Tools

We recommend using **Python** as the primary programming language for this project, with the following key libraries: **Pandas** for data manipulation and analysis, **NLTK** for natural language processing and text preprocessing, and **Scikit-learn** for implementing machine learning algorithms.

6. Submission

Students are required to submit the following components:

1. **Report (PDF Format):**
   - Provide a detailed formalization of the problem, clearly describing each stage of your solution, the features selected, the classifiers used, and the results obtained.
   - Explain in detail the design of each feature, including both content-based and hand-engineered features specific to Arabic text.
   - Present the experimental results, including the evaluation methodology and performance measures such as the confusion matrix, accuracy, precision, recall, and F1-score. Include snapshots of program outputs and visualize results using appropriate graphs. Additionally, provide a comparative analysis of the performance of the different classifiers you trained. For each classifier, clearly report the parameters used. For example, the number of trees and maximum depth for Random Forest, the smoothing parameter for Naïve Bayes, and any other relevant hyperparameters. For the Artificial Neural Network (ANN), describe the **network architecture**, including the number of layers, number of neurons per layer, activation functions, optimization algorithm, learning rate, and other training settings.
   - Discuss any challenges encountered during the project and the strategies used to address them.

2. **Source Code:**
   - Submit all source code developed or extended for this project electronically. No hard copies are required.
   - Ensure that the code is well-documented, properly structured, and runnable. The submission should include all necessary scripts and resources to execute the program successfully.
3. **Demo:**
   - Be prepared to demonstrate your program to the instructor.
   - During the demo, you should be able to operate your program, make minor modifications if requested, and explain and defend the design choices, feature selection, and classification strategies used in your project.

Honor Policy: All are required to adhere to the University honor policy and violations will be dealt with according to university regulations.

Good Luck